

NUMPY: AVERAGE PRODUCT PRICE

AIM

To use NumPy to calculate the average price from a 3×3 array containing sales data for three different products during the past month.

ALGORITHM

1. Start.
2. Import the NumPy library.
3. Create a 3×3 NumPy array. Each row represents a product, and each column represents a recorded product price.
4. Use `numpy.mean()` to calculate the average of all values in the array.
5. Display the average product price.
6. Stop.

PROGRAM (PYTHON)

```
import numpy as np

# Each row represents a product.
# Each column contains a recorded price for that product.
sales_data = np.array([
    [100, 120, 110],
    [200, 180, 220],
    [50, 60, 55]
])

# Calculate the average of all recorded prices
average_price = np.mean(sales_data)

print("Sales data:")
print(sales_data)
print("Average product price:", f"{average_price:.2f}")
```

INPUT

The 3×3 NumPy array (sample prices in currency units):

Record 1	Record 2	Record 3
100	120	110
200	180	220
50	60	55

OUTPUT

Sales data:

```
[[100 120 110]  
 [200 180 220]  
 [ 50  60  55]]
```

Average product price: 121.67

RESULT

The average of all nine recorded price values was calculated using `np.mean()`. The average product price is 121.67 (in the same currency units as the input data).